Milestone 2 Report

1. Introduction

In this milestone, the focus was on building and training an object detection model capable of identifying abnormalities in chest X-ray images. The dataset used was derived from the **VinBigData Chest X-ray Abnormalities Detection dataset**, which contains annotated medical images with bounding boxes indicating abnormal regions.

The goal of this milestone was to:

- Prepare a manageable subset of the dataset
- Convert the dataset into YOLO training format
- Train a deep learning object detection model
- Evaluate the model performance

Due to hardware limitations, only a subset of the full dataset was used to ensure stable training on the available system.

2. Dataset Preparation

The original VinBigData dataset is extremely large (over **140 GB**), making it impractical to download and process entirely on a local machine.

To solve this issue, the following approach was used:

1. The dataset was processed using **Kaggle Notebook**.
2. A subset of **5000 images** was selected from the dataset.
3. DICOM images were converted into **PNG format**.
4. Bounding box annotations were converted into **YOLO format**.

Dataset Structure

The final dataset was structured as follows:

```
dataset
 ├── train
 |    ├── images (4000 images)
 |    └── labels (4000 label files)
 |
 └── val
```

├── images (1000 images)
└── labels (1000 label files)

Each label file contains bounding box coordinates in YOLO format:

class_id x_center y_center width height

3. Data Preprocessing

Several preprocessing steps were performed to prepare the dataset for training:

Image Conversion

Original images were stored in **DICOM format**, which were converted to **PNG images** using a custom script.

Annotation Conversion

Bounding box annotations were converted into YOLO format.

Dataset Splitting

The dataset was split into:

- **Training set:** 4000 images
- **Validation set:** 1000 images

Dataset Verification

A script was used to verify that the number of images and labels matched correctly.

Verification output:

Train images: 4000
Train labels: 4000
Val images: 1000
Val labels: 1000

This confirmed that the dataset was properly prepared for training.

4. Model Selection

For object detection, the **YOLOv8 architecture** from the Ultralytics library was used.

Initially, larger models were considered, but due to system constraints (8 GB RAM and 4 GB GPU VRAM), the **YOLOv8 Nano model (yolov8n)** was selected.

Reasons for choosing YOLOv8n

- Lightweight architecture

- Faster training
- Lower GPU memory usage
- Suitable for systems with limited resources

5. Training Configuration

The model was trained using the following configuration:

Parameter	Value
Model	YOLOv8n
Epochs	40
Image Size	512
Batch Size	2
Training Images	4000
Validation Images	1000
Framework	PyTorch
GPU	NVIDIA RTX 3050

Additional training optimizations included:

- Mixed precision disabled to improve stability
- Image caching disabled due to RAM limitations
- Single worker mode to avoid Windows multiprocessing issues

6. Training Process

Training was executed locally using the Ultralytics YOLOv8 training pipeline.

The model was initialized with **pre-trained weights**, allowing transfer learning to improve training speed and performance.

During training, the following metrics were monitored:

- Bounding box loss
- Classification loss
- Distribution focal loss

- Precision
- Recall
- mAP (Mean Average Precision)

Training completed successfully after **40 epochs**.

Total training time:

~4.8 hours

7. Training Results

Final validation results obtained after training:

Metric	Value
Precision	0.543
Recall	0.248
mAP@0.5	0.308
mAP@0.5:0.95	0.155

Result Interpretation

- **Precision (0.543)** indicates that when the model predicts an abnormality, it is correct about 54% of the time.
- **Recall (0.248)** indicates the model detects approximately 25% of the actual abnormalities present in the images.
- **mAP@0.5 (0.308)** shows moderate detection performance for a lightweight model trained on a limited dataset.

These results demonstrate that the model is capable of detecting abnormalities but still requires improvement through further training and dataset expansion.

8. Model Output

After training, the best model weights were saved as:

best.pt

Location:

runs/detect/train6/weights/best.pt

This model can now be used for:

- Inference on new chest X-ray images
- Model evaluation
- Integration into the CliniScan system

9. Challenges Faced

Several challenges were encountered during this milestone:

1. Dataset Size

The original dataset was too large to download locally.

Solution: A subset of 5000 images was generated using Kaggle.

2. Storage Limitations

Downloading the full dataset required over 140 GB storage.

Solution: A smaller dataset was created and uploaded to Kaggle for easier access.

3. Training Stability

Initial training attempts caused system crashes due to memory usage.

Solution: Training parameters were optimized by:

- Reducing batch size
- Using YOLOv8n model
- Disabling image caching

10. Conclusion

In this milestone, a complete object detection pipeline for chest X-ray abnormality detection was successfully implemented.

Key achievements include:

- Preparing a medical imaging dataset
- Converting annotations to YOLO format
- Training a deep learning detection model

- Evaluating model performance

The trained model demonstrated reasonable detection capability despite hardware limitations and a reduced dataset size.

This milestone establishes the foundation for further improvements in the CliniScan system.

11. Future Work

Future improvements will focus on:

- Training with a larger dataset
- Using higher-capacity YOLO models (YOLOv8s or YOLOv8m)
- Improving detection accuracy
- Implementing explainable AI techniques such as Grad-CAM
- Integrating the trained model into a clinical decision support interface